

63rd ASH Annual Meeting Abstracts

ORAL ABSTRACTS

721.ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT AND ACUTE TOXICITIES

Nutrition As a Predictor of Microbiome Injury in Allo-HCT

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Abstract Background

Intestinal microbiome disruption is a risk factor for poor outcomes after allogeneic hematopoietic cell transplantation (allo-HCT), but the factors that contribute to microbiome injury are not well understood. We hypothesized that nutrition contributes to microbiome composition during allo-HCT.

Methods

Along with 16S profiling of collected fecal samples, we monitored daily inpatient nutritional intake using a customized real-time survey instrument. Data were quality-controlled by a dietitian and matched to the Food and Nutrient Database for Dietary Studies (FNDDS).

Results

97 adult patients received conditioning regimens that were 68% ablative, 22% reduced, and 10% nonmyeloablative; 35% patients had acute myeloid leukemia and 35% had myelodysplastic/myeloproliferative neoplasms, while 10% had non-Hodgkin's lymphoma. Grafts were T-cell depleted in 49% and cord blood in 7%. The remaining had unmodified bone marrow or peripheral blood stem cell. 5 patients had enteral nutrition during the treatment. 22,614 food entries from 5,230 meals were collected during inpatient admissions. Among 800 sequenced stool samples, 329 were collected following exposure to an empiric antibiotic. The hierarchical organization of the FNDDS vocabulary facilitated application of alpha and beta diversities to diet data, as well as analysis of food items (e.g., chicken), which have been reported to more closely associate with microbiome composition than macronutrients (e.g., fat).

Nutritional diversity declined from admission until day 3 (A). Clusters of dietary patterns were revealed by ordination with unweighted UniFrac distance applied, in which highly diverse diets clustered together (B). We observed a correlation between total calories consumed and fecal alpha diversity ($r = 0.23$, $P < 0.001$) and the relative fecal abundance of the genus *Blautia* ($r = 0.31$, $P < 0.001$), which we have previously associated with protection from lethal graft versus host disease (GVHD). In contrast, calorie intake was inversely associated with the fecal relative abundance of genus *Enterococcus* ($r = -0.15$, $P < 0.001$), a genus we have reported exacerbates GVHD. Similar associations with microbiome features were observed for fiber. To gain insight into which types of foods are associated with microbiome injury, we constructed a Bayesian multilevel model to evaluate relationships between microbiota diversity and the amount consumed of different food groups in the two days preceding each stool sample. This model controlled for conditioning intensity, exposure to empiric antibiotics, enteral nutrition, and total

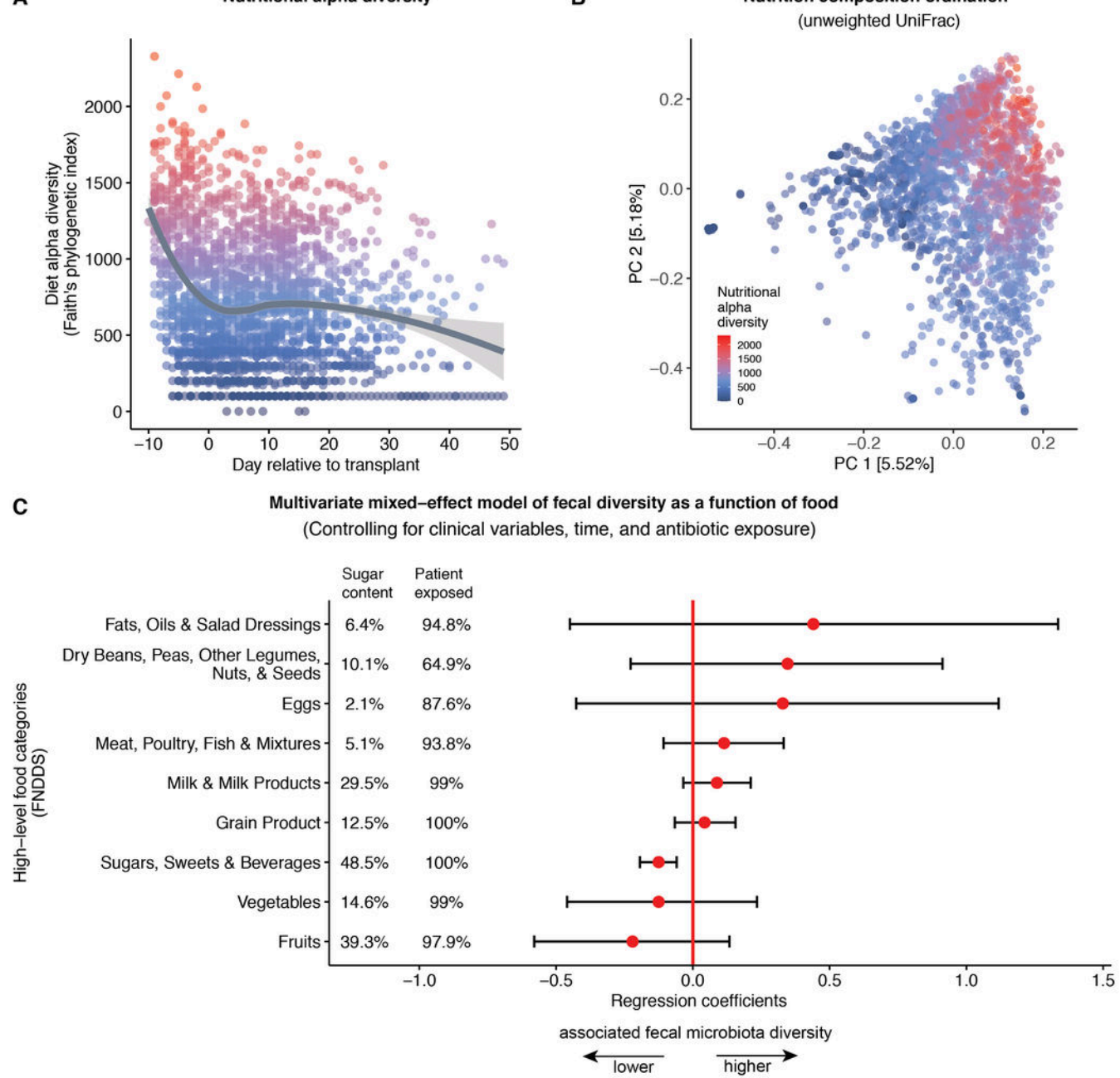
parenteral nutrition. Empiric antibiotics refer to the ones for neutropenic fever such as piperacillin/tazobactam, carbapenems, cefepime, linezolid and for *C. difficile* such as oral vancomycin, and metronidazole. A random intercept term per patient to accommodate repeated measurements from the same patient and a random intercept term for each week relative to HCT were incorporated in the model. Intake of sugars, sweets and beverages was associated with low fecal microbiota alpha diversity (C). The model predicts that, on average, consumption of 100g sugars and sweets over two days would result in a biologically meaningful decline of diversity by 1.13-fold in inverse Simpson units. Interestingly, fruits, a food type enriched in simple sugars, trended toward associations with lower diversity as well.

Conclusion

Consumption of sugars and sweets is associated with lower fecal microbiota alpha diversity in allo-HCT. We hypothesize that initial insults to diverse microbial communities are exacerbated by simple sugars, which can be exploited by the remaining organisms as readily available nutrients. These results highlight the importance of developing evidence-based nutritional recommendations in allo-HCT.

Disclosures Adintori: Vidafuel Inc.: Current holder of stock options in a privately-held company. Buchan: Savor Health: Current Employment. Gomes: Xbiome: Current Employment. Johnson: Diversigen: Consultancy. Knights: Diversigen: Consultancy. Jenq: Microbiome DX: Consultancy; Merck: Consultancy; Prolacta: Consultancy, Membership on an entity's Board of Directors or advisory committees; Kaleido: Consultancy, Current holder of stock options in a privately-held company, Membership on an entity's Board of Directors or advisory committees, Research Funding; Seres: Consultancy, Current holder of stock options in a privately-held company, Membership on an entity's Board of Directors or advisory committees, Patents & Royalties, Research Funding; LisCure: Consultancy, Membership on an entity's Board of Directors or advisory committees; MaaT Pharma: Consultancy, Membership on an entity's Board of Directors or advisory committees; Karius: Consultancy. Giralt: GSK: Membership on an entity's Board of Directors or advisory committees; PFIZER: Membership on an entity's Board of Directors or advisory committees; SANOFI: Membership on an entity's Board of Directors or advisory committees; JAZZ: Membership on an entity's Board of Directors or advisory committees; AMGEN: Membership on an entity's Board of Directors or advisory committees; JENSEN: Membership on an entity's Board of Directors or advisory committees; BMS: Membership on an entity's Board of Directors or advisory committees; CELGENE: Membership on an entity's Board of Directors or advisory committees; Actinnum: Membership on an entity's Board of Directors or advisory committees. Perales: NexImmune: Honoraria; Servier: Honoraria; MorphoSys: Honoraria; Novartis: Honoraria, Other; Nektar Therapeutics: Honoraria, Other; Miltenyi Biotec: Honoraria, Other; Merck: Honoraria; Medigene: Honoraria; Takeda: Honoraria; Sellas Life Sciences: Honoraria; Kite/Gilead: Honoraria, Other; Karyopharm: Honoraria; Incyte: Honoraria, Other; Equilium: Honoraria; Cidara: Honoraria; Celgene: Honoraria; Bristol-Myers Squibb: Honoraria; Omeros: Honoraria. van den Brink: Priothera: Research Funding; Da Volterra: Other: has consulted, received honorarium from or participated in advisory boards; Jazz Pharmaceuticals: Honoraria; Pharmacyclics: Other; Seres: Other: Honorarium, Intellectual Property Rights, Research Funding and Stock Options; Wolters Kluwer: Patents & Royalties; Pluto Therapeutics: Current holder of stock options in a privately-held company, Other: has consulted, received honorarium from or participated in advisory boards; Amgen: Honoraria; Frazier Healthcare Partners: Honoraria; Forty-Seven, Inc.: Honoraria; Notch Therapeutics: Honoraria; Nektar Therapeutics: Honoraria; GlaskoSmithKline: Other: has consulted, received honorarium from or participated in advisory boards; Kite Pharmaceuticals: Other; Novartis (Spouse): Other: has consulted, received honorarium from or participated in advisory boards; Juno Therapeutics: Other; Magenta Therapeutics: Honoraria; Ceramedix: Other: has consulted, received honorarium from or participated in advisory boards; Rheos: Honoraria; DKMS (nonprofit): Other; Therakos: Honoraria; Merck & Co, Inc: Honoraria; SyntheKine (Spouse): Other: has consulted, received honorarium from or participated in advisory boards; Lygenesis: Other: has consulted, received honorarium from or participated in advisory boards; WindMIL Therapeutics: Honoraria. Schluter: Postbiotics Plus LLC: Other: cofounder. Peled: MaaT Pharma: Consultancy; CSL Behring: Consultancy; DaVolterra: Consultancy; Seres Therapeutics: Research Funding.

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A. Diet diversity declines during allo-HCT. Each point is a single patient's day of nutritional intake. **B.** Different dietary compositions are observed in allo-HCT. Each point is one patient's nutritional intake on a single day. Similar diets are clustered together in this principal coordinate analysis. Highly diverse diets cluster together. **C.** Low fecal microbiota alpha diversity is associated with consumption of sugars, sweets and beverages, and more broadly, of foods rich in simple sugars. Plotted are estimates for the association between food types and fecal diversity as regression coefficients of posterior distributions from a Bayesian multilevel model; whiskers are 95% intervals. The values of the horizontal axis convey the expected amount of diversity change (in natural log inverse Simpson units) following exposure to each 100 grams of the indicated food in the prior two days.

Figure 1